A-A-54441
15 July 1991

COMMERCIAL ITEM DESCRIPTION

CEMENT, ZINC POLYCARBOXYLATE, Dental

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-C-36871 which is cancelled.

This commercial item description covers a zinc polycarboxylate cement suitable for use as a luting agent for crowns, bridges, inlays, space maintainers and orthodontic bands.

Salient characteristics:

Except as noted below, the cement shall be in accordance with ANSI/ADA Specification No. 61. In the case of a conflict, this document shall govern.

Material. The zinc polycarboxylate cement shall be composed essentially of a zinc oxide-magnesium oxide powder and polycarboxylate copolymer (consisting of copolymer of acrylic acid and one or more polycarboxylic acids) liquid. The zinc oxide and magnesium oxide shall be in accordance with the requirements of the Pharmacopeia of the United States of America (U.S.P.). Any ingredients used in the cement which are not included in the U.S.P. or National Formulary (N.F.) shall be of high quality throughout and shall be suitable for the purpose intended.

Components. The zinc polycarboxylate dental cement shall consist of the following: one bottle containing a minimum of 90 grams of powder, one bottle containing a minimum of 50 milliliters of liquid, an eye dropper, one measuring scoop and one mixing pad. The powder shall not contain any fillers such as quartz. The cement shall be packaged in a container designed to prevent damage during storage or shipment.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Physical properties. In addition to ADA Specification No. 61, the cement shall meet the following requirements:

Working time (minutes)	1 1/2 (minimum)
Setting time at 37°C (min.)	5-9
at 25°C (min.)	8-15
Compressive strength (psi)	8000-9600
Diametral tensile strength (psi)	500-1300
Color	light beige or buff

Delivery schedule. No more than 2 months shall have elapsed from the date of manufacturer of the cement to the date of delivery to the Government.

Toxicity. The manufacturer shall submit a statement that the cement has no toxic effects on a normal, healthy person when used for the purpose intended.

Adhesion. The cement shall adhere to both the tooth and the button when tested as specified herein.

Instructions. In addition to the requirements of ADA Specification No. 61, the instructions shall include information on preparation of the tooth surface prior to application of the cement.

Workmanship. The cement shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Package (PG). One package containing one bottle of powder, one bottle of liquid, one scoop, one mixing pad, and one set of instructions, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. T he government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted with MIL-STD-105 at an inspection level S-2 with an AQL of 2.5 (percent defective). The unit of product for test purposes shall be one package of cement.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 at an inspection level of S-1, with an AQL of 1.0 (percent defective). The unit of product for test purposes shall be one package of cement.

Tests.

Diametral tensile strength. Prepare the samples in the same manner as for the compressive strength. Measure the diameter and length of each sample prior to testing.

Pad each side of the sample with polyethylene 1 to 2 mils (10^{-3} inch) thick. Load the sample at a crosshead speed of 0.02 inch per minute until the sample fractures. Calculate the tensile strength using the following formula:

$$\text{Tensile strength} = \frac{2P}{3.14\,DL}$$

Where P is the load in pounds at fracture
D is the diameter of the sample in inches
L is the length of the sample in inches

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Adhesion. A human tooth and a stainless steel orthodontic lingual button shall be used. The tooth shall be prepared in accordance with standard dental procedures prior to applying the cement. The button shall be attached to the tooth and kept moist and stored at 37.0° ± 0.5°C for three hours. The tooth and button shall then be separated.

The adhesion test is not required on all lots provided the following conditions are met. The first lot or batch passes the adhesion test. The contractor provides certification that the lots or batches being offered to the government are of the same formulation and manufacturing process as the first lot.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specification, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6520-00-141-0420	CEMENT, ZINC POLYCARBOXYLATE, Dental

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project # 6520-2394
Location: ML #1/CEMZINC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I. RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54441	2. DOCUMENT DATE (YYMMDD) [illegible]91

3. DOCUMENT TITLE
CEMENT, ZINC, POLYCARBOXYLATE, Dental

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Defense Personnel Support Center	(215)737-2870	444-2870
c. ADDRESS (Include Zip Code) ATTN: DPSC-MSE 2800 S. 20th Street Philadelphia, PA 19101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403 Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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